

tier2_cuttlefish_ab

tier2_cuttlefish_ab.sh — user guide

Field	Value
Revision	1
Last modified	2026-06-11T00:00:00Z
Status	active — host-gated (SKIPs cleanly off a Linux+KVM host)
Authority	Helix OTA control-plane / device-integration team
Related	docs/design/CUTTLEFISH_TIER2.md (the bring-up plan + sources), docs/design/EMULATED_DEVICE_TESTING.md (Tier-1/2/3 overview), tests/emulator/ab_virt/ (the planned A/B-virt PWU-AB-3 this PWU-CF-2 case mirrors)

Overview

tests/emulator/tier2_cuttlefish_ab.sh exercises the **real Android update_engine A/B + AVB/dm-verity + auto-rollback** flow on a Cuttlefish (cvd) virtual Android device — the hardware-free fidelity tier the dev-host QEMU A/B-virt tier (tests/emulator/ab_virt/) cannot reach, and the closest proxy for the Orange Pi 5 Max / RK3588 OTA apply.

It runs in two phases on a Linux + nested-KVM host:

1. **Slot-flip phase** — launch_cvd → read baseline active slot → apply an OTA payload to the inactive slot via update_engine → reboot → assert the active slot **FLIPPED** + dm-verity is present on the new slot.
2. **Auto-rollback phase (PWU-CF-2, headline)** — corrupt the now-INACTIVE previous slot (mark it unbootable / fail dm-verity), reboot, and assert the device **AUTO-ROLLS-BACK** to the known-good active slot (active slot does not revert to the corrupted one; boot succeeds on the good slot; the rollback trace is captured). This mirrors the planned ab_virt PWU-AB-3 corrupt-slot rollback case.

Honest status (§11.4.6 / §11.4.123)

The entire Linux flow — including the PWU-CF-2 auto-rollback section — is **UNVERIFIED-pending-Linux-host**. It is implemented from the latest official AOSP docs (see docs/design/CUTTLEFISH_TIER2.md “Sources verified”), but has **not** been executed on a real Linux+KVM host. The Helix dev host is macOS on Apple Silicon (applehv hypervisor, no /dev/kvm), so the script **SKIPS cleanly there** (exit 3, topology_unsupported) — never a fake PASS.

The auto-rollback section does **not** claim Android auto-rollback works until the script is run on the real host. The two AOSP-UNCONFIRMED: steps it depends on — the exact OTA-apply driver, and the exact safe corrupt-the-inactive-slot mechanism (legacy A/B direct write vs Virtual A/B COW/snapuserd path; bootctl availability) — are **detected + attempted at runtime and FAIL honestly** if they do not reproduce (§11.4.6 no-guessing, §11.4.123 rock-solid-proof-or-research, §11.4.133 verified-before-destructive-write).

Prerequisites

- **Topology (mandatory):** Debian-based Linux + usable /dev/kvm (nested-virt on a cloud instance, an M4+/macOS-15 nested-virt Mac, or a Linux box). On any host without it, the script SKIPS (exit 3). Verify per CUTTLEFISH_TIER2.md §3: `grep -c -w "vmx\|svm" /proc/cpuinfo` (x86) or `find /dev -name kvm` (ARM64).
- **Cuttlefish host packages:** cuttlefish-base installed (run --prepare once, then reboot — see Usage). Group membership in kvm, cvdnetwork, render.
- **Tooling:** git, apt/dpkg, python3, adb/fetch_cvd (from the cvd host package), ~30 GB disk, network (AOSP build fetch).

Usage examples

```
# One-time prerequisite install (builds + installs the cuttlefish
  debs; needs sudo),
# then REBOOT to load kernel modules + udev rules:
tests/emulator/tier2_cuttlefish_ab.sh --prepare
sudo reboot

# Normal run (Linux+KVM host) – runs the full slot-flip + auto-
  rollback flow:
tests/emulator/tier2_cuttlefish_ab.sh

# On this macOS dev host – honest SKIP (the only thing verifiable
  here):
tests/emulator/tier2_cuttlefish_ab.sh # -> [SKIP] ...
  topology_unsupported ; exit 3
```

Environment

Var	Default	Meaning
HELIX_CF_DIR	./cuttlefish	cuttlefish workdir (debs, cvd images)
HELIX_CF_TARGET	auto by arch (aosp_cf_arm64_only_phone / aosp_cf_x86_64_only_phone)	the cuttlefish build target
HELIX_CF_EVIDENCE	docs/qa/<run-id>/ cuttlefish_ab/	captured-evidence dir (§11.4.69/ §11.4.83)

Inputs / Outputs / Side-effects

- **Inputs:** topology (Linux+KVM), cuttlefish host packages, an OTA payload (ota.zip + update_device.py) for the apply phase.
- **Outputs:** captured evidence under the evidence dir — kvm.txt, getprop_before.txt, update_engine.txt, slot_after.txt, verity.txt, virtual_ab.txt, corrupt_bootctl.txt / corrupt_dd.txt, corrupt_setactive.txt, slot_after_rollback.txt, rollback_trace.txt, rollback_logcat.txt, transcript.txt — plus a PASS / FAIL / SKIP verdict on stdout.
- **Exit codes:** 0 = PASS (real flow ran green on a Linux+KVM host), 1 = FAIL (a real defect, or an UNVERIFIED-pending-host step that did not reproduce — never a fake PASS), 3 = SKIP (topology absent / prerequisites missing).
- **Side-effects:** boots a cvd virtual device; the auto-rollback phase performs a **bounded** corruption of the **inactive** slot only (one 4K block, or a bootctl set-slot-as-unbootable on the inactive index) — never the active/good slot (§11.4.133). Self-cleaning: stop_cvd runs on every exit path (trap ... EXIT INT TERM, §11.4.14).

Edge cases

- **No /dev/kvm (e.g. macOS):** SKIP exit 3, topology_unsupported. Verified on the macOS dev host.
- **cuttlefish-base not installed:** SKIP exit 3 with a pointer to --prepare.
- **fetch_cvd / OTA package absent:** the apply phase FAILs honestly (UNVERIFIED-pending-host), it does **not** fake a PASS.
- **bootctl shell command absent:** the corrupt phase falls back to a bounded dm-verity-fail write on the inactive system<slot> by-name partition; if neither is available it FAILs honestly (no silent skip-to-PASS).
- **Device hangs after corruption:** treated as FAIL (a hang is **not** a rollback PASS); the rollback_trace.txt is captured for host-side forensics.
- **Auto-rollback does not occur (boots the corrupted slot):** FAIL, not PASS.

Internal behaviour

Topology gate (Linux + readable/writable /dev/kvm) → arch→target selection → prerequisite/--prepare check → fetch+launch cvd → baseline slot → real update_engine apply → reboot + slot-flip assertion → dm-verity assertion → **PWU-CF-2**: detect A/B variant (ro.virtual_ab.enabled) → corrupt the inactive slot (bootctl set-slot-as-unbootable, else bounded dm-verity-fail write) → set the corrupted slot active to force the bad-boot path → reboot → assert boot succeeds on the known-good slot + capture the rollback trace. All adb/curl/loop waits are bounded; every PASS cites an evidence path (§11.4.69 ab_pass_with_evidence shape).

Related scripts

- tests/emulator/ab_virt/ — the dev-host QEMU A/B-virt tier whose PWU-AB-3 corrupt-slot rollback case this PWU-CF-2 mirrors.
- docs/design/CUTTLEFISH_TIER2.md — the bring-up plan, sourced FACTs, and the UNCONFIRMED: verify-before-run register (§8) this script honours at runtime.

Last verified

- macOS dev host SKIP path (exit 3, topology_unsupported): verified 2026-06-11.
- sh -n + bash -n parse-clean (§11.4.67): verified 2026-06-11.
- Linux+KVM full flow (slot-flip + auto-rollback): **UNVERIFIED-pending-Linux-host** — to be run when the operator's Linux+KVM host is attached.